

CURBCOUNT · REQUIREMENTS

CurbCount requirements

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DOCUMENT CONTROL

Document ID	CBC-REQ-001	Version	0.6
Status	Draft	Date	2026-10-01
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/curbcount		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-09-30	Amish Chadha	Status figures from CBC-CAL-001 v0.3 after the design for construction (CBC-DDR-003); no status changed
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from CBC-CAL-001; design case bike lane 2.0 m to match R6; decisions from CBC-DDR-001 reflected; proposed restatements of R4 and R11 noted
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (numbered, measurable requirements with targets and concept status)
0.1	2026-09-25	Amish Chadha	Initial scaffold

CurbCount requirements

These requirements are targets for concept review and must be revised with a co-design partner before the design is frozen. The design choices behind them (thermal array, tracking on FieldNode's STM32WL with standard FieldNode power, 15-minute bins and three classes in a 10-byte record, jumper-only calibration, public notice) were decided by Amish on 2026-09-25 (CBC-DDR-001 and CBC-DDR-002). Version 0.4 applies his decisions: R4 is restated in pixels per meter at head height, R7 is restated for temperate sites, R13 follows the new \$275 value-engineering target, and R11 stays at 6 kg. Version 0.5 takes its figures from CBC-CAL-001 v0.4, after the design for construction (CBC-DDR-003); no status changed. Status is taken from the calculation note CBC-CAL-001 v0.4; "met on paper" means met by calculation only. No requirement is **not met**. Three are at risk (R1, R2, R7) and four cannot be shown at TRL 3 (R3, R10, R12, R14).

Design case: one counter on a 114 mm pole 0.45 m behind the curb, sensor window 4.3 m above the road, a 3 m sidewalk, a 2.0 m bike lane and one 3.5 m traffic lane.

Table 1. Requirements.

ID	Requirement	Target	Verification	Status (CBC-CAL-001 v0.4)
R1	Count pedestrians, cyclists (including micromobility) and motor vehicles in the nearest lane, by direction along the street	3 classes x 2 directions	Test against manual counts	At risk: 4.2 frames for a car at 50 km/h, 4 assumed needed
R2	Count accuracy for pedestrians and cyclists	Within ± 10 % of manual counts per hour at flows up to 600 per hour per direction	Field comparison with manual counts	At risk: random merging alone loses 6.5 %; groups add more
R3	Count accuracy for motor vehicles in the nearest lane	Within ± 15 % of manual counts per hour	Field comparison	Not verifiable at TRL 3 (4.2 frames, 3.4 px smear)
R4	Privacy: no image, audio or identifier leaves the device; frames held only in RAM; sensor unable to resolve faces or plates	8 px/m or less at head height for a 1.75 m person, below the IEC 62676-4 detection level of 25 px/m (restated in v0.4; was a ground pixel of 0.25 m or larger)	Design review, firmware audit, geometry	Met on paper: 7.6 px/m (8.5 px/m for a 2.0 m person)
R5	Output format	15-minute counts per class and direction in an open, documented 10-byte record (six 12-bit counters and a status byte) over LoRaWAN	Design review	Met by design
R6	Coverage from one pole	Full width of a 3 m sidewalk, a 2 m bike lane and the nearest lane	Geometry, field check	Met on paper with the array turned 90 degrees: -4.45 to 8.34 m across
R7	Operating range	Survive -20 °C to +50 °C; count at temperate sites (design days with air up to 20 °C). Hot-climate sites use the mmWave radar variant (restated in v0.4; was counting with air up to 35 °C)	Thermal chamber, seasonal field test	At risk: on the mild design day sunlit pavement hides pedestrians for 10.2 h with single frames, 3.2 h with track averaging; shaded 0 h
R8	Autonomy without sun	3 days of continuous counting	Energy budget, bench test	Met on paper: 6.22 days on one cell (4.36 at -20 °C)
R9	Energy neutral in winter	At 1.5 peak sun hours with street shading derating	Energy budget, field log	Met on paper: FieldNode's 6 W panel stores 4.4 Wh/day against 2.47 Wh/day
R10	Ingress protection	IP65 for the enclosure and the sensor head	Spray test	Not verifiable at TRL 3
R11	Mass on the pole	6 kg or less	Weighing	Met on paper: 5.26 kg with the parts added for construction (4.44 kg in v0.4; 6.02 kg with the 20 W panel)
R12	Installation	Two trained people, 60 min, no drilling, welding or pole wiring; withstands 35 m/s gusts	Timed trial, bracket calculation	Not verifiable at TRL 3: 59 min estimate; wind and fit met on paper

ID	Requirement	Target	Verification	Status (CBC-CAL-001 v0.4)
R13	Parts cost	\$275 or less per counter (value-engineering target, budget_usd ; was \$150)	Priced BOM	Within the value-engineering target: \$274.00 (\$1.00 under)
R14	Service life	5 years outdoors with one battery change	Design review, UV-stable materials	Not verifiable at TRL 3
R15	Radio use	Within EU868 1 % duty cycle and the US915 dwell limits	Airtime calculation	Met on paper: 0.82 s/h at SF9; 371 ms at US915 DR0

Table 2. Assumptions behind the targets.

Assumption	Value	Source or basis
Walking speed	1.0 to 1.5 m/s	Typical design range (estimate)
Cyclist speed	up to 25 km/h	Typical urban maximum (estimate)
Vehicle speed in the nearest lane	up to 50 km/h	Common urban limit (estimate)
Peak sun hours, winter design case	1.5 h	Mid-latitude winter (estimate)
Street shading and dust derating	40 %	Estimate, to be checked on real poles
Temperate design day	Air 10 to 20 °C	CBC-CAL-001 section D